

CAP-MADELEINE, QC, Canada

WMO: 714250

Lat: 49.2508N

Lon: 65.3247W

Elev: 95

StdP: 14.65

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.2	-1.4	-13.9	2.6	-4.5	-9.9	3.2	-0.9	40.4	15.8	36.4	13.7	17.0	270	0.759

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	74.9	64.9	71.7	63.0	69.1	61.6	66.6	72.8	64.6	70.0	62.8	67.9	11.5	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.1	89.8	70.6	62.1	83.5	68.1	60.4	78.7	66.3	31.4	72.9	29.8	70.1	28.5	68.0	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
33.9	29.7	26.5	-9.9	83.6	4.6	3.5	-13.2	86.1	-15.9	88.1	-18.5	90.1	-21.9	92.6
			-10.7	70.7	4.7	1.7	-14.1	72.0	-16.9	73.0	-19.5	73.9	-22.9	75.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	39.5	15.9	16.3	24.4	35.3	45.6	55.4	62.0	61.1	53.9	44.1	34.2	23.9
	DBStd	17.60	9.70	9.61	8.85	6.36	6.06	6.11	4.95	5.05	6.00	6.17	7.36	8.01
	HDD50	4942	1059	945	794	441	169	17	0	0	25	206	476	810
	HDD65	9379	1524	1365	1259	890	603	293	121	143	335	648	923	1275
	CDD50	1096	0	0	0	1	31	180	372	343	143	23	3	0
	CDD65	55	0	0	0	0	0	5	27	21	2	0	0	0
	CDH74	141	0	0	0	1	1	20	63	49	7	0	0	0
	CDH80	18	0	0	0	0	0	3	8	5	2	0	0	0
Wind	WSAvg	12.3	15.6	14.6	13.9	13.0	10.8	9.8	9.1	9.3	10.5	12.3	13.7	15.4
Precipitation	PrecAvg	35.3	2.8	2.3	3.0	2.5	2.4	2.7	3.3	3.4	2.6	2.8	2.5	4.1
	PrecMax	47.4	6.7	7.9	8.2	6.1	5.7	4.7	8.2	6.6	6.2	6.3	7.2	8.6
	PrecMin	27.2	0.9	0.7	1.1	0.8	0.7	1.1	1.1	1.5	0.9	0.6	0.7	1.2
	PrecStd	6.1	1.5	1.5	1.5	1.4	1.1	0.9	1.6	1.2	1.3	1.6	1.3	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.5	44.5	48.7	59.3	69.9	76.9	80.0	78.9	73.8	67.4	62.5	48.0
		MCWB	44.7	41.0	42.0	51.5	56.5	65.4	66.7	66.9	65.0	57.8	57.3	43.5
	2%	DB	37.7	35.9	42.5	51.0	62.6	71.0	74.9	74.4	69.2	61.4	53.1	41.2
		MCWB	34.6	33.4	37.4	43.4	53.1	61.0	65.5	64.9	61.3	54.4	47.8	38.3
	5%	DB	32.9	32.3	38.4	46.2	57.6	66.9	71.6	70.9	65.7	56.2	47.2	37.7
		MCWB	31.2	30.5	34.2	40.4	49.7	58.4	63.4	63.1	59.2	51.0	42.8	35.1
	10%	DB	29.4	29.7	35.0	42.9	54.0	63.9	69.0	68.2	62.5	52.7	43.4	34.4
		MCWB	27.7	28.1	31.8	38.1	47.5	56.6	62.1	61.7	56.9	48.1	39.8	32.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.9	41.5	44.0	52.3	59.3	67.5	70.0	69.1	66.4	59.8	56.8	44.4
		MCDB	48.4	44.8	48.4	58.8	66.8	74.5	76.8	76.9	72.0	65.4	62.5	47.6
	2%	WB	34.8	34.2	38.0	44.1	54.5	63.2	67.1	66.1	62.7	54.8	47.7	38.6
		MCDB	37.0	35.6	41.4	49.0	60.9	69.2	72.9	72.5	67.6	60.5	52.0	40.5
	5%	WB	31.0	30.9	34.7	40.9	50.9	60.0	64.8	64.2	59.8	51.3	43.0	35.3
		MCDB	32.6	32.2	38.0	45.5	56.5	65.3	70.1	69.5	64.5	55.8	46.1	37.3
	10%	WB	27.6	28.6	32.4	38.6	48.4	57.7	63.0	62.4	57.4	48.4	40.3	32.5
		MCDB	29.0	29.9	34.8	42.3	53.3	63.0	67.9	67.3	61.9	52.0	42.7	34.0
Mean Daily Temperature Range	5% DB	MDBR	12.2	12.0	11.4	10.4	11.8	12.4	11.4	10.8	11.1	9.6	9.2	9.5
		MCDBR	18.9	14.5	15.1	16.8	18.5	17.2	15.5	14.9	16.5	16.2	15.7	14.8
		MCWBR	18.2	13.1	11.3	11.8	11.6	10.3	9.3	9.3	11.3	12.2	13.5	13.5
	5% WB	MCDBR	18.9	13.2	13.8	15.6	17.4	15.7	14.3	13.7	15.5	15.8	15.1	13.8
		MCWBR	18.0	12.3	11.1	11.7	11.9	10.7	9.6	9.3	11.4	12.4	13.3	12.7
Clear-Sky Solar Irradiance	taub	0.267	0.276	0.301	0.369	0.362	0.380	0.416	0.380	0.357	0.336	0.296	0.272	
	taud	2.372	2.307	2.313	2.255	2.413	2.377	2.326	2.423	2.459	2.495	2.515	2.416	
	Ebn at Noon	254	279	289	278	285	279	266	272	268	254	243	236	
	Edh at Noon	21	28	33	39	35	37	38	33	29	24	18	17	
All-Sky Solar Radiation	RadAvg	318	618	1017	1393	1635	1801	1725	1497	1125	689	347	236	
	RadStd	31	56	87	113	104	148	156	80	84	48	39	25	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (16 neighbors)	N/A	N/A	+2.38	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page